Tokenization is the process of breaking text into smaller units called tokens. A token can be a word, part of a word, punctuation mark, or another piece of text, depending on the tokenizer being used.

Artificial intelligence systems process language differently from humans. When a sentence such as 'The quick brown fox jumps over the lazy dog.' is given to a language model, the text is first transformed into tokens before further processing.

RAG stands for Retrieval-Augmented Generation. In a RAG system, documents are loaded, split into chunks, converted into useful representations, and retrieved when a user asks a question. Good document processing can make retrieval more accurate.

Here is some random text for practice: blue clouds move slowly across the morning sky while seven small birds fly between two old trees. A yellow bicycle rests near a wooden fence, and someone has left a notebook on a bench.

Numbers and symbols can also be useful for tokenization experiments: 123, 4567, 3.14159, 2026, email@example.com, #learning, AI/RAG, and Python_101.

Try observing how different tokenizers handle contractions, punctuation, repeated words, and unusual expressions. For example: don't, can't, we're, hello!!!, machine-learning, tokenization... and RAG-based systems.

Learning by experimenting is often more useful than simply reading documentation. Change the text, load the PDF, inspect the extracted content, and then count or display the tokens produced by your tokenizer.